

Digital Forensics

Ex.No:2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by TestDisk:

- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

☒ If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

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The first partition is listed twice which points to a corrupted partition or an invalid partition table entry. Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem. Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

TestDisk displays the first results in real time. (click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder (click on thumb).

After the Deeper Search, the results are displayed as follows: The first partition "**Partition 1**" was found by using backup boot sector. In the last line of your display, you can read the message "**NTFS found using backup sector!**" and the size of your partition. The "partition 2" is displayed twice with different size. **Partitions listed as D(eleted) will not be recovered** if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.

- Highlight the first partition Partition 2 and press **p** to list its data.

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The file system of the upper logical partition (label Partition 2) is damaged (click on thumb).

- Press q for Quit to go back to the previous display.
- Let this partition Partition 2 with a damaged file system marked as D(deleted).
- Highlight the second partition Partition 2 below
- Press p to list its files.

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from D(eleted) to **L(ogical)**. This way you will be able to recover this partition.

Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **P**Primary partition.

- Press Enter to proceed.

Partition table recovery

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It's now possible to write the new partition structure. **Note:** The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named Partition 1 is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.

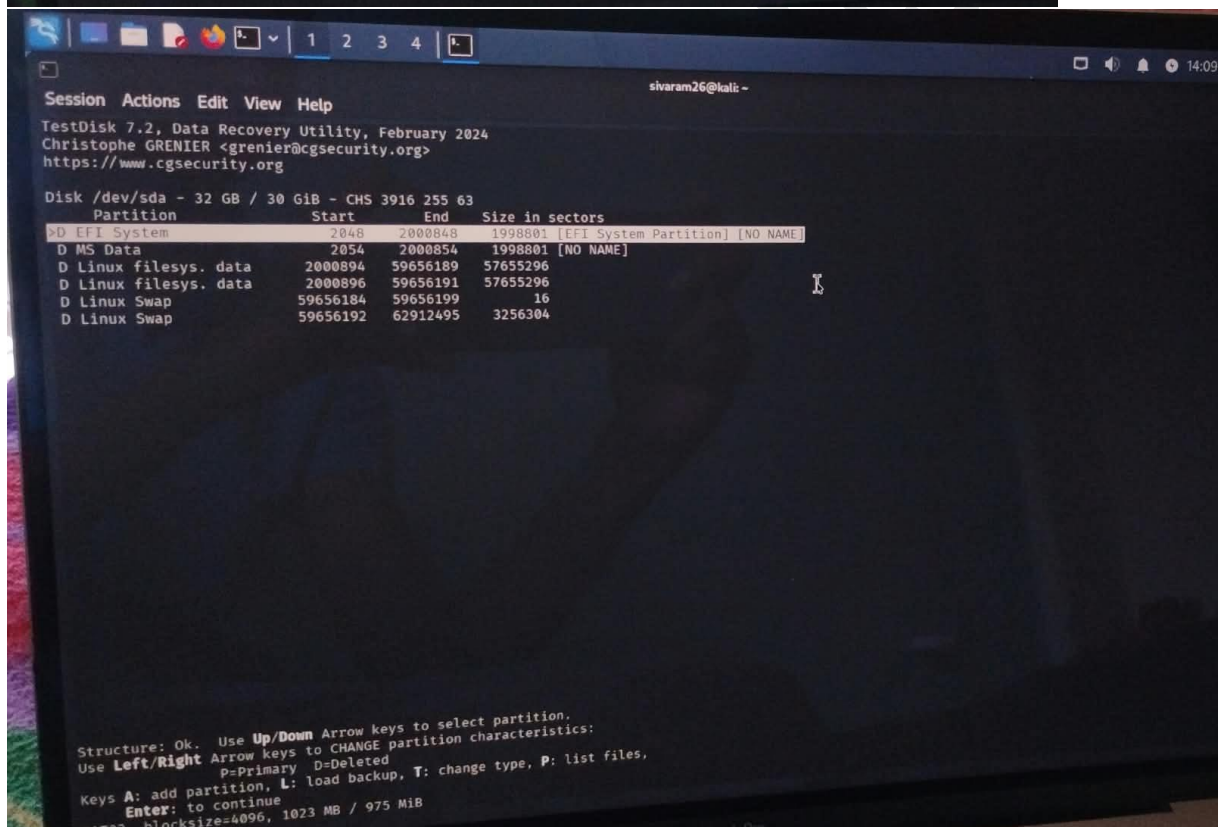
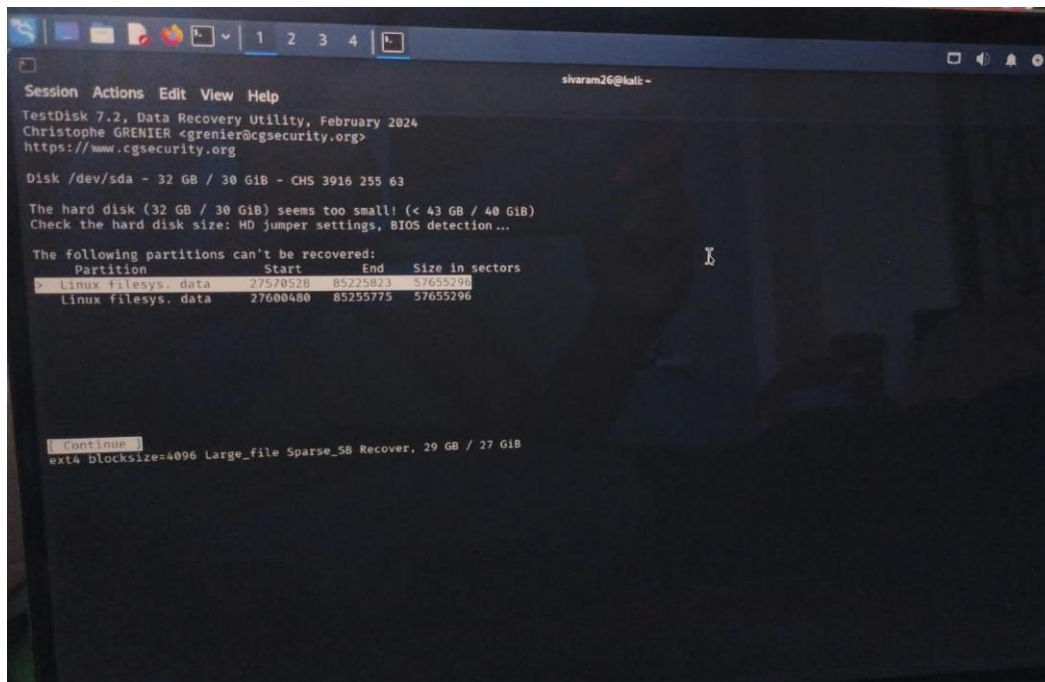
- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use y to confirm and next OK.

More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:

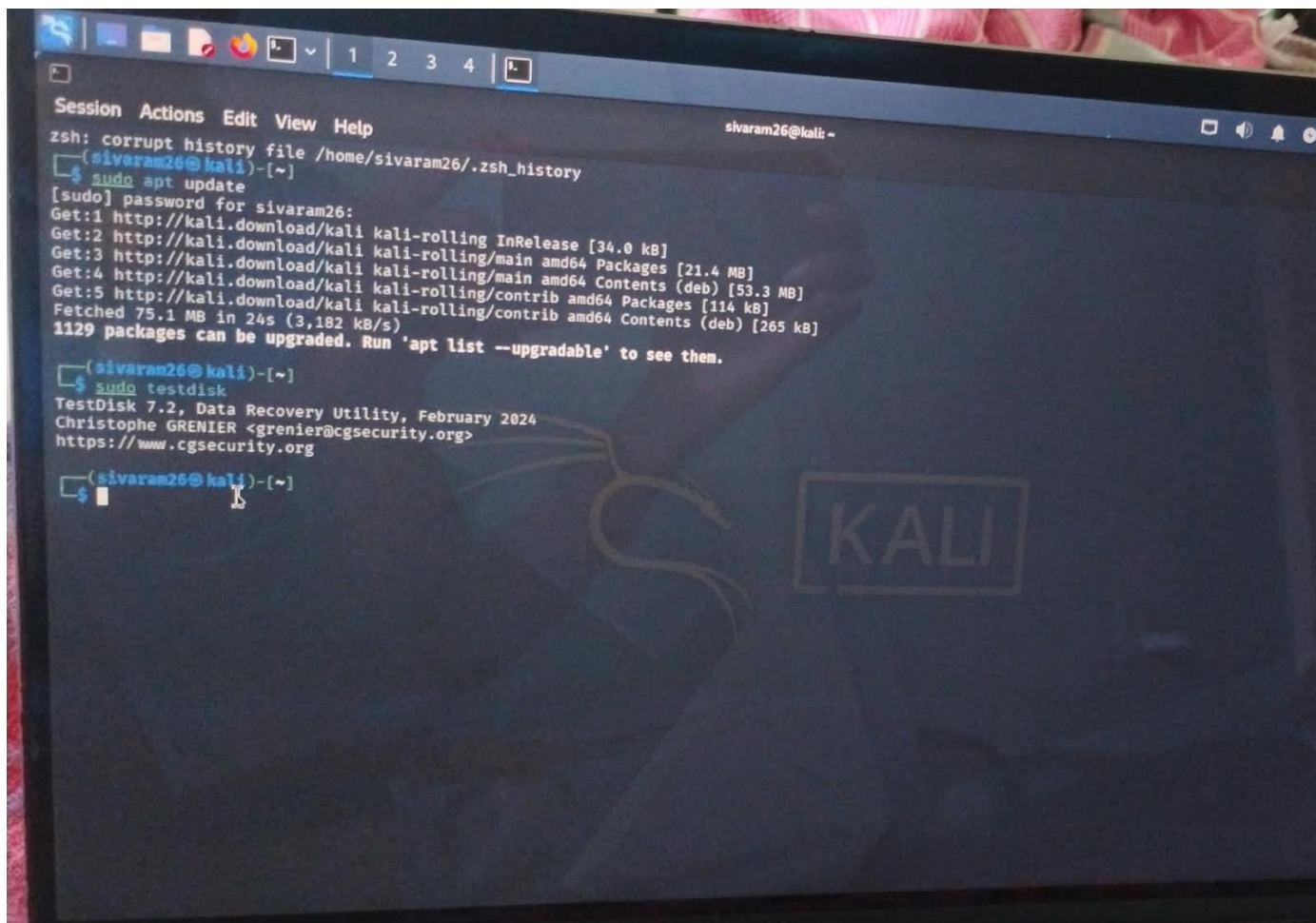
The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.
- TestDisk displays **You have to restart your Computer to access your data** so press Enter a last time and reboot your computer.

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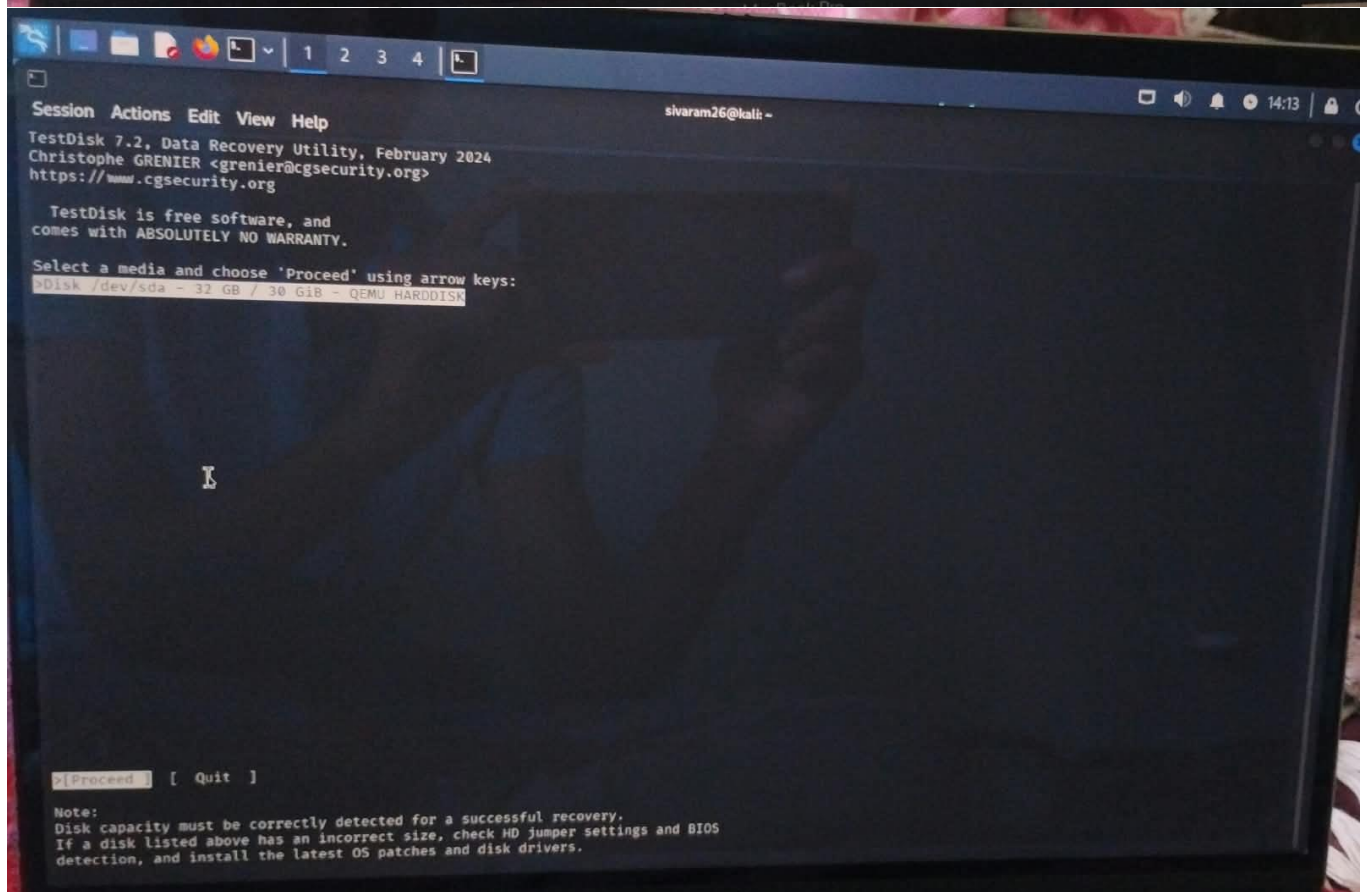


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A terminal window on a Kali Linux system. The user runs 'zsh: corrupt history file /home/sivaram26/.zsh_history'. Then they run '\$ sudo apt update', which shows progress for updating Kali rolling repositories. After the update, they run '\$ sudo testdisk', which displays the TestDisk 7.2 splash screen with copyright information for Christophe GRENIER.

```
Session Actions Edit View Help
sivaram26@kali: ~
zsh: corrupt history file /home/sivaram26/.zsh_history
(sivaram26@kali)~$ sudo apt update
[sudo] password for sivaram26:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.3 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [265 kB]
Fetched 75.1 MB in 24s (3,182 kB/s)
1129 packages can be upgraded. Run 'apt list --upgradable' to see them.
(sivaram26@kali)~$ sudo testdisk
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
(sivaram26@kali)~$
```



The same terminal window showing the TestDisk main menu. It prompts the user to select a media and choose 'Proceed' using arrow keys. The first option is 'Disk /dev/sda - 32 GB / 30 GiB - QEMU HARDDISK'. At the bottom, there are '[Proceed]' and '[Quit]' options, along with a note about disk capacity detection.

```
Session Actions Edit View Help
sivaram26@kali: ~
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk /dev/sda - 32 GB / 30 GiB - QEMU HARDDISK

[Proceed] [Quit]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```